Master's Thesis Progress Documentation

Phase 3: Data Preprocessing

1. Thesis Title

Design and Evaluation of an Interactive Visual Analytics System for Explainable Artificial Intelligence

2. Purpose of This Document

This document summarizes the work completed during Phase 3 of the master's thesis.

The main purpose of this phase was to prepare the cleaned IBM Telco Customer Churn dataset for machine learning. The preprocessing process included removing the identifier column from the model input, encoding the target variable, transforming categorical variables, scaling continuous numerical variables, splitting the dataset, and saving the processed outputs.

This phase builds directly on the Exploratory Data Analysis completed in Phase 2. At the end of Phase 2, the dataset contained 7,032 complete customer records and was ready for preprocessing.

The processed datasets created in this phase will be used during the next phase for training and evaluating machine learning models.

3. Current Stage of the Thesis

The thesis has now completed the dataset exploration and preprocessing stages.

The research concept was defined during Phase 1, while Phase 2 focused on dataset selection, data cleaning, and Exploratory Data Analysis.

Phase 3 focused on transforming the cleaned dataset into a numerical format that can be used by machine learning algorithms.

This preprocessing stage follows the methodology described in the thesis exposé, including categorical encoding, feature scaling, and the removal of irrelevant variables.

4. Work Completed in Phase 3

4.1 Loading the Cleaned Dataset

The original IBM Telco Customer Churn dataset was loaded into a new Google Colab notebook named:

02_preprocessing.ipynb

The same cleaning steps used in Phase 2 were applied again to ensure that the preprocessing notebook is reproducible and can run independently.

The TotalCharges column was converted from text into a numerical data type. The 11 incomplete records identified during Phase 2 were removed.

The final cleaned dataset contained:

- 7,032 customer records
- 21 original columns
- No missing values

4.2 Feature Group Definition

The dataset columns were separated into different groups according to their role in the preprocessing process.

The target variable was:

- Churn

The identifier column was:

- customerID

The numerical features were:

- SeniorCitizen
- tenure
- MonthlyCharges
- TotalCharges

The continuous numerical features were:

- tenure
- MonthlyCharges
- TotalCharges

The binary numerical feature was:

- SeniorCitizen

The categorical features were:

- gender
- Partner
- Dependents
- PhoneService
- MultipleLines

- InternetService
- OnlineSecurity
- OnlineBackup
- DeviceProtection
- TechSupport
- StreamingTV
- StreamingMovies
- Contract
- PaperlessBilling
- PaymentMethod

In total, 19 input features were selected for machine learning.

4.3 Removal of the Identifier Column

The customerID column contains a unique value for every customer.

This column was not used as an input feature because it does not describe customer behaviour and does not provide meaningful information for churn prediction.

However, the customer IDs were preserved separately. They may later be used to connect model predictions, SHAP explanations, and Power BI records to individual customers.

4.4 Separation of Features and Target

The dataset was separated into:

- Input features represented by X
- Target variable represented by y
- Customer identifiers stored separately

The input feature dataset contained:

- 7,032 rows
- 19 original input features

The target variable contained the churn status for each customer.

4.5 Target Variable Encoding

The Churn target variable originally contained text values.

The values were converted into numerical form using the following mapping:

- No = 0
- Yes = 1

After encoding, the dataset contained:

- 5,163 non-churn customers
- 1,869 churn customers

This confirmed that the target variable was imbalanced, with fewer churn customers than non-churn customers.

The imbalance will be considered during the machine learning model evaluation phase.

4.6 Train-Test Split

The dataset was divided into training and testing sets using an 80–20 split.

The resulting datasets contained:

- 5,625 training records
- 1,407 testing records

A random state of 42 was used to make the split reproducible.

Stratification was also applied using the target variable. This ensured that the percentage of churn and non-churn customers remained similar in both the training and testing datasets.

The training dataset contained:

- 1,495 churn customers

The testing dataset contained:

- 374 churn customers

4.7 Continuous Feature Scaling

The continuous numerical features were standardized using StandardScaler.

The scaled features were:

- tenure
- MonthlyCharges
- TotalCharges

Standardization transforms the variables so that they have an average close to zero and a standard deviation close to one.

This helps machine learning algorithms such as Logistic Regression compare variables that originally use different measurement scales.

The scaler was fitted only on the training dataset. The fitted scaler was then applied to the testing dataset.

This was done to prevent information from the testing data from influencing the training process.

4.8 Binary Feature Handling

The SeniorCitizen feature was already stored in binary form:

- 0 represents a non-senior customer
- 1 represents a senior customer

Since the variable was already numerical and binary, it was preserved in its original form and was not scaled.

4.9 Categorical Variable Encoding

The 15 categorical variables were converted into numerical form using one-hot encoding.

One-hot encoding creates separate binary columns for the different categories within each feature.

For example, a feature such as Contract can be represented through separate columns for:

- Month-to-month
- One year
- Two year

The encoder was configured to ignore unknown categories. This prevents errors when a category appears in future data but was not present in the training dataset.

The one-hot encoder was fitted only on the training data and then applied to the testing data.

4.10 Preprocessing Pipeline

A ColumnTransformer was used to combine the preprocessing steps into one structured pipeline.

The transformer applied:

- StandardScaler to continuous numerical features
- No transformation to the binary SeniorCitizen feature
- OneHotEncoder to categorical features

Using one preprocessing pipeline improves consistency and reduces the chance of applying different transformations to the training and testing datasets.

4.11 Processed Dataset Creation

After applying the preprocessing pipeline, the original 19 input features were transformed into 45 numerical model features.

The processed training dataset contained:

- 5,625 rows
- 45 numerical features

The processed testing dataset contained:

- 1,407 rows
- 45 numerical features

Readable feature names were created for all transformed columns.

The processed arrays were then converted into Pandas DataFrames so that they could be inspected and saved more easily.

4.12 Dataset Validation

Several validation checks were performed after preprocessing.

The checks confirmed that:

- The processed training dataset contained no missing values.
- The processed testing dataset contained no missing values.
- No infinite values were present.
- Training and testing datasets contained the same feature columns.
- All processed model inputs were numerical.
- The number of training and testing targets matched the corresponding feature datasets.
- Customer IDs remained correctly aligned with the original records.

These checks confirmed that the processed datasets were suitable for machine learning.

4.13 Prevention of Data Leakage

Data leakage occurs when information from the testing dataset influences the model during training.

To prevent this problem:

- The train-test split was completed before fitting the preprocessing transformer.
- The scaler was fitted only on the training data.
- The one-hot encoder was fitted only on the training data.
- The fitted preprocessing pipeline was then applied to the testing data.

This approach ensures that the testing dataset remains independent and can later provide a fair evaluation of model performance.

4.14 Reference Dataset Creation

Training and testing reference datasets were also created.

These files preserve the original customer information, including:

- Customer IDs
- Original feature values
- Original churn values
- Encoded churn values

The reference datasets will be useful during later phases when model predictions and SHAP explanation values need to be connected back to individual customers.

They will also support the integration of prediction and explanation results into Power BI.

4.15 Saving the Processed Outputs

All preprocessing outputs were stored inside the following folder:

phase3_outputs

The following files were created:

- cleaned_telco_churn.csv
- X_train_processed.csv
- X_test_processed.csv
- y_train.csv
- y_test.csv
- train_reference.csv
- test_reference.csv
- feature_name_mapping.csv
- preprocessor.joblib
- preprocessing_metadata.json

The preprocessor.joblib file stores the fitted preprocessing pipeline.

The metadata file records the feature lists, target mapping, train-test split settings, and preprocessing methods.

These files improve the reproducibility of the project and will be used in the following phases.

5. Tools and Libraries Used

The following tools and libraries were used during Phase 3:

- Google Colab

- Python
- Pandas
- NumPy
- Scikit-learn
- Joblib
- JSON
- Google Colab file utilities

Scikit-learn was mainly used for:

- Train-test splitting
- Standardization
- One-hot encoding
- Column-based transformations

6. Key Preprocessing Decisions

The main preprocessing decisions were:

- Remove customerID from machine learning inputs.
- Preserve customerID for later prediction tracking.
- Encode Churn as No = 0 and Yes = 1.
- Use an 80–20 train-test split.
- Apply stratification based on the target variable.
- Scale continuous numerical variables.
- Preserve SeniorCitizen as a binary feature.
- Apply one-hot encoding to categorical variables.
- Fit all preprocessing transformations only on training data.
- Save the fitted preprocessing pipeline for future use.

These decisions provide a clear and reproducible foundation for model development.

7. Key Outcomes

At the end of Phase 3:

- The cleaned dataset was successfully loaded.
- The identifier column was removed from the machine learning input.
- The target variable was converted into numerical format.
- The dataset was divided into training and testing sets.
- Target stratification was successfully applied.
- Continuous numerical variables were standardized.
- Categorical variables were one-hot encoded.
- The original 19 features were converted into 45 model-ready features.
- No missing or infinite values remained.
- Training and testing columns were consistent.
- Data leakage prevention was applied.
- Reference datasets were created.
- The fitted preprocessing pipeline was saved.

- The processed data is now ready for machine learning model development.

8. Next Steps

The next phase of the thesis will focus on machine learning model development.

The planned tasks include:

1. Load the processed training and testing datasets.
2. Train several classification models.
3. Develop a Logistic Regression model.
4. Develop a Decision Tree model.
5. Develop a Random Forest model.
6. Consider a gradient boosting model such as XGBoost.
7. Evaluate the models using suitable classification metrics.
8. Compare model performance.
9. Examine the effect of class imbalance.
10. Select the most suitable model for SHAP explainability analysis.
11. Save the selected model for the next phase.

The main evaluation metrics will include:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion matrix

9. Summary

Phase 3 focused on preparing the IBM Telco Customer Churn dataset for machine learning.

The cleaned dataset was separated into training and testing datasets, the target variable was encoded, continuous variables were standardized, and categorical variables were transformed using one-hot encoding.

The preprocessing pipeline was fitted only on the training data to avoid data leakage. The final processed datasets contained 45 numerical model features and passed all validation checks.

The required datasets, preprocessing pipeline, metadata, and reference files were saved successfully.

The project is now ready to continue with Phase 4, which will focus on training, evaluating, and comparing machine learning classification models.